NewAE at CASCADE 2026

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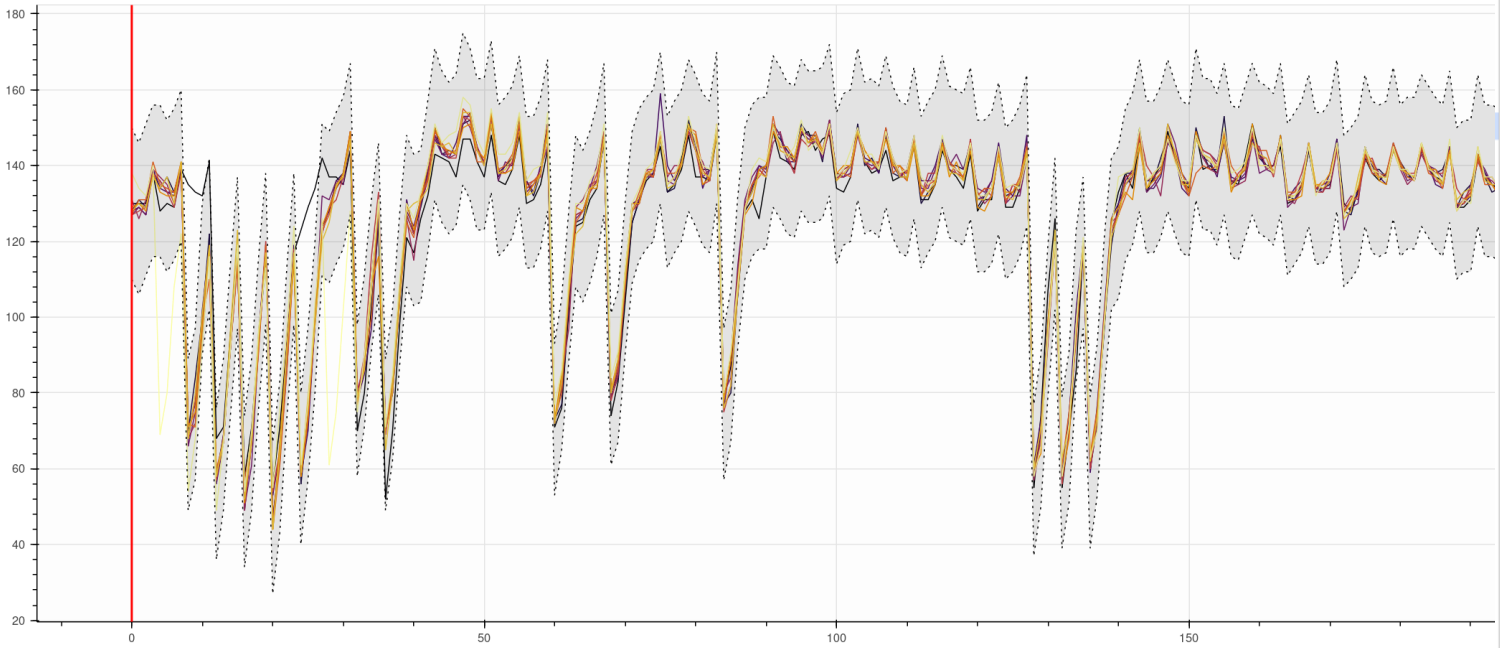
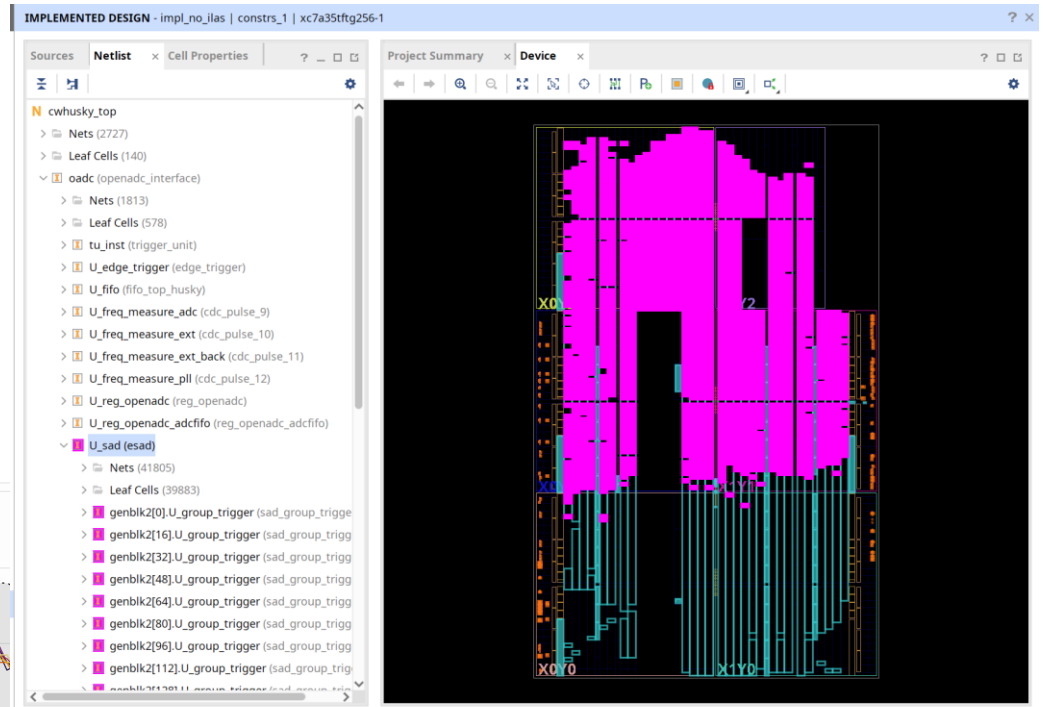
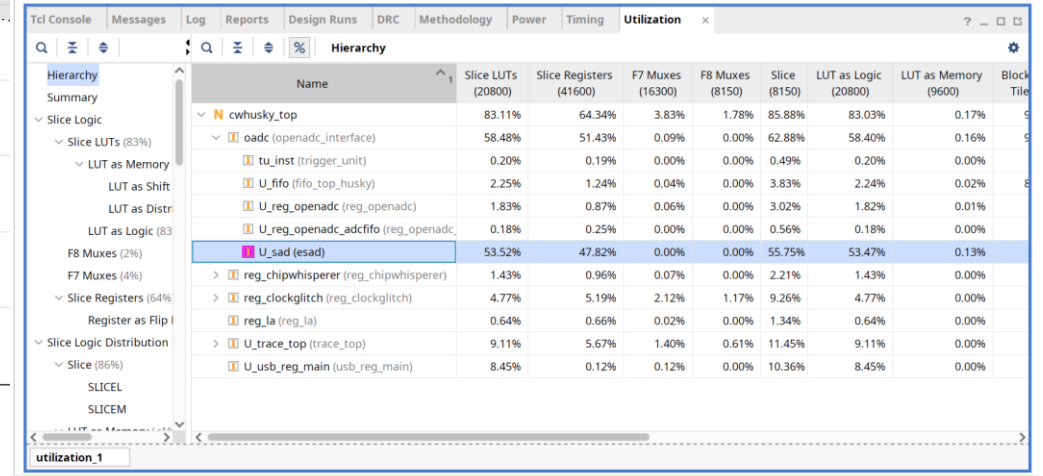
NewAE Technology

Demos: SAD Triggering

- **S**um of **A**bsolute **D**ifferences
- Capabilities approaching state-of-the-art
- 512 samples at 250 MS/s
- Shockingly expensive in FPGA resources



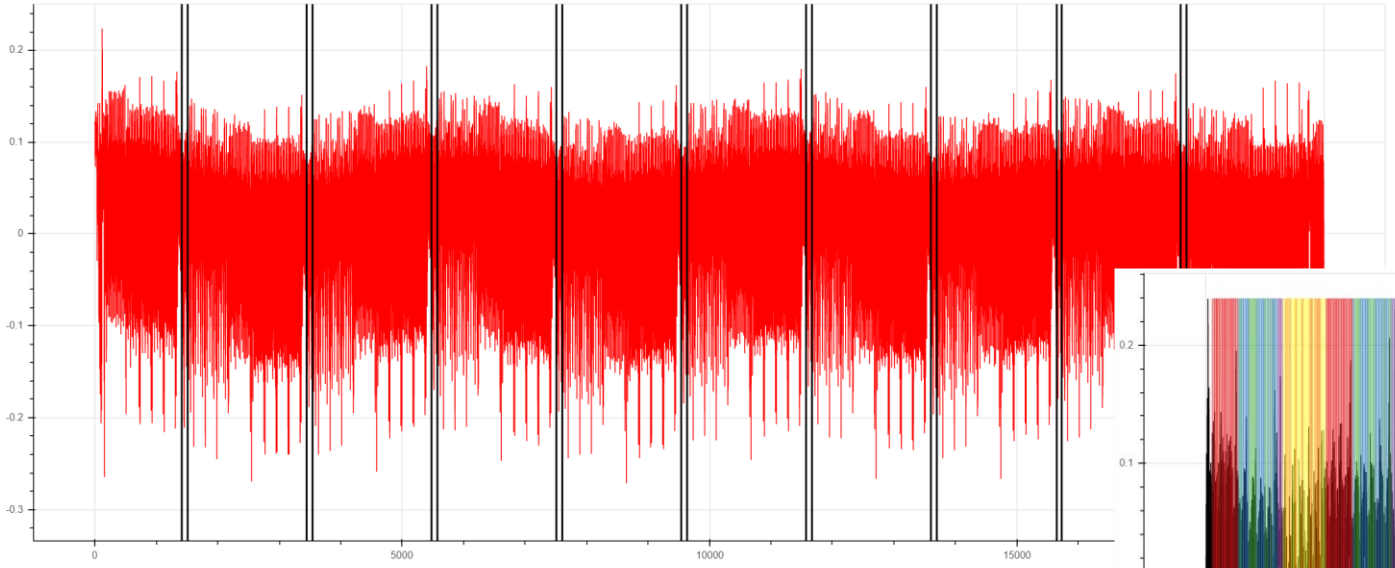


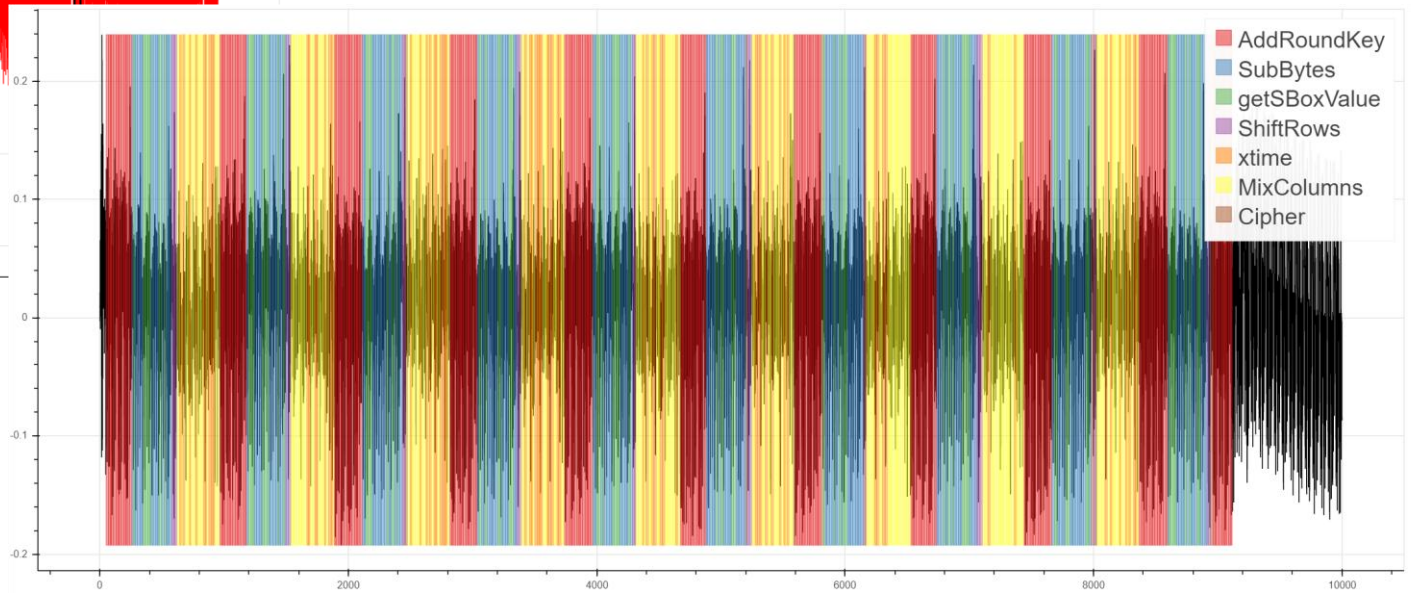


Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	F8 Muxes (8150)	Slice (8150)	LUT as Logic (20800)	LUT as Memory (9600)	Block Tile
cwhusky_top	83.11%	64.34%	3.83%	1.78%	85.88%	83.03%	0.17%	9
oadc (openadc_interface)	58.48%	51.43%	0.09%	0.00%	62.88%	58.40%	0.16%	9
tu_inst (trigger_unit)	0.20%	0.19%	0.00%	0.00%	0.49%	0.20%	0.00%	
U_fifo (fifo_top_husky)	2.25%	1.24%	0.04%	0.00%	3.83%	2.24%	0.02%	8
U_reg_openadc (reg_openadc)	1.83%	0.87%	0.06%	0.00%	3.02%	1.82%	0.01%	
U_reg_openadc_adcfifo (reg_openadc_	0.18%	0.25%	0.00%	0.00%	0.56%	0.18%	0.00%	
U_sad (esad)	53.52%	47.82%	0.00%	0.00%	55.75%	53.47%	0.13%	
reg_chipwhisperer (reg_chipwhisperer)	1.43%	0.96%	0.07%	0.00%	2.21%	1.43%	0.00%	
reg_clockglitch (reg_clockglitch)	4.77%	5.19%	2.12%	1.17%	9.26%	4.77%	0.00%	
reg_la (reg_la)	0.64%	0.66%	0.02%	0.00%	1.34%	0.64%	0.00%	
U_trace_top (trace_top)	9.11%	5.67%	1.40%	0.61%	11.45%	9.11%	0.00%	
U_usb_reg_main (usb_reg_main)	8.45%	0.12%	0.12%	0.00%	10.36%	8.45%	0.00%	

Demos: TraceWhisperer

- Complex Arm debug technology
- Tons of potential applications for side-channel work





New Things Coming Soon

New Online Course

- Revamp with focus on ChipWhisperer Husky
- New CTF style labs
- Higher quality lessons/labs

New Project/Analyzer

- Simplified project/analyzer interface
- Big speed increase (orders of magnitude)

Hardware Bit-Banging Serial Interfaces

- I2C, 1-wire, SWD, and more
- Precise FPGA-controlled timing
- Use it to trigger:
  - Power traces
  - Glitches